A Project Tittle

On

Customer Segmentation Using K-Means and Hierarchical Clustering

Machine Learning

Bachelor of technology

In

A.Y.:2026-27

By

Name: SURAPURAJU HIMASREE **Roll**: 2520090181

Name: BONALA SUDHARSHINI **Roll**:2520090227

Under the Esteemed guidance of

Dr. Lakshmi Prasanna Yeluri

Asst Professor, Professor-in-charge

CSIT

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd u/s 3 of the UGC Act, 1956)

Off Campus: Bachupally Gandimaisamma Road, Bowrampet, Hyderabad, Telangana 500 043

Phone No. 7815926816, www.klh.edu.in

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DEPARTMENT OF CSIT

Name of Title: Customer Segmentation Using K-Means and Hierarchical Clustering

1.Abstract:

Customer segmentation is an important technique that helps businesses understand their customers by grouping them based on similar characteristics and behaviors. Instead of treating every customer the same, companies can identify different customer groups and provide products, services, and marketing strategies that better match their needs. This project focuses on customer segmentation using two popular unsupervised machine learning techniques: K-Means Clustering and Hierarchical Clustering.

K-Means Clustering divides customers into a predefined number of groups by finding patterns in the data. Hierarchical Clustering, on the other hand, creates a tree-like structure of customer groups, allowing different levels of similarity to be explored. By comparing these two approaches, the project aims to identify meaningful customer segments and understand how each method organizes data.

The project uses customer information such as age, income, spending habits, or purchase history to discover natural groupings without relying on pre-labeled data. Basic evaluation methods are used to determine the quality of the clusters and select an appropriate number of groups. The identified customer segments can then be analyzed to understand their common characteristics and preferences.

The results of this project can help businesses improve customer satisfaction by offering personalized recommendations, targeted promotions, and better marketing strategies. It also supports more informed business decisions by identifying valuable customer groups and understanding purchasing behavior.

Overall, this project demonstrates how clustering techniques can transform raw customer data into meaningful insights. By applying K-Means and Hierarchical Clustering, organizations can better understand their customers, improve business planning, and enhance customer engagement. The project provides a practical introduction to unsupervised machine learning while highlighting its usefulness in solving real-world business problems through effective customer segmentation.

2. INTRODUCTION:

Customer segmentation is the process of dividing customers into groups based on similar characteristics such as age, income, purchasing behavior, and spending patterns. It helps businesses better understand their customers and create personalized marketing strategies, improve customer satisfaction, and make informed business decisions. Instead of treating every customer the same, companies can focus on the specific needs of different customer groups.

This project focuses on customer segmentation using two popular unsupervised machine learning techniques: K-Means Clustering and Hierarchical Clustering. These methods identify natural patterns in customer data without requiring pre-labeled information. K-Means groups customers into a fixed number of clusters based on similarity, while Hierarchical Clustering creates a tree-like structure that shows relationships between different customer groups. Comparing these techniques helps in understanding their effectiveness for customer analysis.

The main objective of this project is to identify meaningful customer segments that can help businesses improve marketing strategies, recommend suitable products, and enhance customer engagement. By applying clustering techniques to customer data, the project demonstrates how machine learning can transform raw data into useful insights and support better business decision-making.

3. SOFTWARE REQUIREMNETS:

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook or Visual Studio Code
- **Libraries Used:**
  - NumPy – Numerical computations
  - Pandas – Data manipulation and analysis
  - Matplotlib – Data visualization
  - Seaborn – Statistical data visualization
  - Scikit-learn – Machine learning algorithms (K-Means and Hierarchical Clustering)
  - SciPy – Hierarchical clustering and dendrogram generation
- **Dataset:** Customer Segmentation Dataset (CSV format)

Signature of the Guide   Course Instructor